

WORK EXPERIENCE

RiVer Lab, Northeastern University

Boston, MA

Robotics Research Engineer (Prof. Taskin Padiş)

May 2026 – Present

- Leading development of a real-time pedestrian detection and tracking system for an autonomous wheelchair; deployed YOLOv8n fused with Orbbec RGB-D depth data to estimate pedestrian position and velocity in unstructured outdoor environments.
- Implementing localization using ArUco markers, IMU, and wheel encoders to precisely estimate wheelchair pose on public sidewalks.

Human-Robot Interaction Lab, Northeastern University

Boston, MA

Robotics Research Engineer (Prof. Zhi Tan)

October 2025 – April 2026

- Designed and deployed a full autonomy pipeline from MuJoCo simulation to physical Stretch 3 for assistive HRI research; implemented a hierarchical macro-action system coordinating navigation, perception, and grasping as a unified mobile manipulation stack.
- Achieved reliable autonomous pick-and-place on real hardware using Nav2, LiDAR SLAM (AMCL), ArUco-based object detection, and a Jacobian IK solver; integrated speech recognition for voice-commanded task execution, demonstrated end-to-end in a 6-minute video.
- Architected a production-grade ROS2 codebase with modular components ; surfaced and minimized failure modes including localization drift and contact dynamics. [\[Deployment Repo\]](#) [\[Simulation Repo\]](#)

Rivian (via RV-Tech)

Palo Alto, CA

Embedded Software Engineer (Intern)

June 2025 - August 2025

- Pioneered an automated diagnostics system for embedded vehicle software, analyzing sensor data and system logs to detect regressions during CI and release testing. Eliminated manual triage and accelerated release validation.
- Deployed a root-cause analysis pipeline on AWS with LLM integration to correlate sensor recordings with connectivity logs.
- Integrated Databricks to centralize diagnostic telemetry and automate reporting across embedded programs and HIL systems.

Samsung Research America - Think Tank Team

Mountain View, CA

Embedded Systems Engineer

August 2023 - December 2023

- Prototyped an AI assistant earbud by designing a PCB around a Qualcomm processor and flashing a custom RTOS on it.
- Built a Voice Activation Detection algorithm using FFT in Python and C; integrated it into the REX RTOS to reduce the latency.
- Designed a bidirectional voice pipeline under strict latency constraints to enable real-time conversational interaction.

Amazon Robotics

North Reading, MA

Robotics Software Engineer (Intern)

May 2022 - August 2022

- Developed a digital-twin of an Amazon warehouse station in Isaac Sim, simulating robots with rigid-body-dynamics and photorealism.
- Eliminated all collision risks and operational costs of training facility by transitioning operator onboarding to simulation workflows.
- Extended Isaac Sim's source code by implementing VR integration, including custom 3D transformation functions and input interfaces.
- Engineered high-fidelity robot trajectory generation using trapezoidal velocity profiles and S-curves with jerk limits.

Tufts University

Medford, MA

Research Engineer

December 2023 - May 2024

- Integrated LiDAR into embedded research platforms, developed STM32 firmware for real-time data acquisition; mentored 2 researchers

Research Engineer

2021 - December 2023

- Built and deployed an automated sensor fabrication device for Tufts NanoLab, enabling on-demand production of tension sensors.
- Accelerated prototyping cycles from weeks to hours and ensured reproducible sensor characteristics critical for research.
- Designed an interrupt-based stepper motor controller and a real-time PID controller to regulate sensor tension in embedded C.
- Presented the project at [MRS-Boston \(Tufts News\)](#); published it as first author at MDPI ([Publication Link](#))

SELECT PROJECTS

ROS2 Multi-Sensor Perception Simulation in Isaac Sim [\(Project Link\)](#)

- Built a multi-sensor perception simulation in Isaac Sim 5.0 featuring a 128-channel RTX LiDAR and stereo RGBD camera, streaming synchronized PointCloud2, Image, Odometry, and TF data to RViz2 via ROS2 bridge with Action Graph architecture.
- Implemented domain randomization over lighting (500–3000 lux) and LiDAR noise parameters; produced a systematic sim-to-real gap analysis documenting beam visibility artifacts, physics model limitations, and sensor rate discrepancies — directly informing hardware sensor selection and validation strategy.

Custom 3D Software Rendering & Physics Simulation Engine [\(Project Link\)](#)

- Built a C++ software rendering and real-time physics engine from scratch without external libraries, prioritizing full determinism and performance control.
- Implemented real-time buoyancy and torque simulation via dynamic force integration; engineered depth buffering, triangle rasterization with shading, and back-face culling from first principles.
- Designed a complete 3D projection pipeline — matrix transforms, OBJ model loading, pixel buffer masking — without any graphics API; handled multi-threaded window resizing and frame-rate limiting for stable timing under dynamic load.

Incremental 3D Point Cloud Reconstruction (SfM) [\(Project Link\)](#)

- Built an incremental Structure-from-Motion pipeline using SIFT feature detection, RANSAC-based Essential Matrix estimation, and PnP camera registration to reconstruct a colored sparse 3D point cloud from monocular images; minimized reprojection error via bundle adjustment and exported georeferenced PLY output with full camera trajectory recovery.

Servo Motor Feedback Controller [\(Project Link\)](#)

- Designed and built an open-source closed-loop DC motor driver that converts DC motors into precision servos; implemented a PID controller with sub-50 μ s latency and trapezoidal trajectory planning in C on STM32, operating at 5–45 Volts and 4.1 A supply range.
- Integrated current sensing for torque feedback with high-frequency filtering and cyclic buffering enabling torque feedback; fabricated a PCB integrating H-bridge, quadrature encoder interface, and UART/I2C, validated under sensor noise and fault conditions.
- Released as a complete MIT-licensed hardware product with production-ready Gerber files, pre-compiled firmware, assembly video, and full user guide; supports daisy-chaining up to 112 units on a single I2C bus via Qwiic connectors across 5 versioned releases.
- Authored a 16-page technical manuscript with experimental validation (step response, trajectory tracking under load), BOM analysis, and manufacturing workflow; submitted to *HardwareX* [\(Manuscript\)](#)

TECHNICAL SKILLS

- **Software & Control:** C/C++, Python, MATLAB, RTOS, ROS2, PID/Impedance Control, Git/GitLab, Inverse Kinematics
- **Robotics:** PyTorch, RL, MuJoCo, Isaac Sim, SLAM, GTSAM, Kalman Filter, Factor Graphs, OpenCV, Sensor Fusion, YOLOv8
- **Hardware & Embedded:** STM32, ESP32, PCB Design (Altium, KiCad), Motor Drivers (Brushed/Brushless), Sensor Integration (Encoders, Force Sensors, IMUs), I2C, SPI, UART, CAN

EDUCATION

Northeastern University

Boston, MA

Master of Science in Robotics

2024 - May 2026

- Reinforcement Learning, Feedback Control Systems, Geometric Deep Learning, Autonomous Field Robotics (3D SLAM, Kalman Filters, Sensor Fusion, GTSAM, Factor Graphs), Legged Robotics (Bipedal locomotion)

Tufts University

Medford, MA

Bachelor of Science in Electrical and Computer Engineering

2019-2023

- Related Courses: Algorithms, Computational Geometry, Data structures and C++, Embedded systems, electronics 2, Control Systems, probabilistic system analysis, Parallel Computing(Cuda++)